MIL-C-16154E
6 MAY 1966
SUPERSEDING
MIL-C-0016154D(SHIPS)
15 JUNE 1964
MIL-C-16154C
18 SEPTEMBER 1953
(SEE 6.3)

MILITARY SPECIFICATION

COMPUTER INDICATOR, RADIAC CP-95( )/PD (READER FOR RADIAC DETECTOR DT-60( )/PD)

The specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a Radiac Computer Indicator CP-95( )/PD which is capable of computing and indicating the amount of X and gamma radiation to which the Radiac Detector DT-60( )/PD has been exposed.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-S-901 — Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment and Systems, Requirements for.

MIL-D-16153 — Detector, Radiac DT-60( )/PD.

MIL-E-16400 — Electronic Equipment, Naval Ship and Shore, General Specification.

MIL-E-17555 — Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.

MIL-F-19207/16 — Fuseholders, Types FHN26G and FHN26W.

MIL-R-22732 — Reliability Requirements for Shipboard and Ground Electronic Equipment.

STANDARDS

MILITARY

MIL-STD-108 — Definitions of and Basic Requirements

FSC 6665

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for Enclosures for Electric and Electronic Equipment

MIL–STD–454 — Standard General Requirements for Electronic Equipment.

PUBLICATIONS

MILITARY

NAVSHIPS 93820 — Handbook for prediction of Shipboard and Shore Electronic Equipment Reliability.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Applications for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York.)

3. REQUIREMENTS

3.1 Preproduction sample. Prior to beginning production a sample shall be tested as specified in 4.3 (see 6.2).

3.2 General. The equipment shall be in accordance with MIL–E–16400, except as otherwise specified herein.

3.3 Reliability. The following reliability requirements as specified in MIL–R–22732 shall be applicable to this equipment. The mean-time-between failures (MTBF) of the equipment is predicted to be 5,000 hours using method D of NAVSHIPS 93820.

3.3.1 The contractor shall conduct the reliability assurance program that is applicable to a preproduction and production model.

3.3.2 *Level.* The reliability requirement for this equipment shall be 300 hours, mean-time-between-failure (MTBF).

3.3.3 Reliability evaluation shall be performed in accordance with the procedures for "Reliability Assurance by analysis and prediction", as specified in MIL–R–22732.

3.3.4 *Testing.* Quantitative reliability requirements shall be demonstrated as specified in 4.2.2.

3.3.5 In addition to the regular reliability reports of MIL–R–22732, reliability prediction reports shall be submitted whenever the design changes or additional design data becomes available affecting the prediction.

3.4 Design requirements.

3.4.1 *Equipment composition.* The computer-indicator shall consist of a 3,650 angstrom near ultraviolet-light source, a photosensitive element, an amplifier if required for satisfactory compliance with the requirements of this specification, an indicating meter, detector positioning assembly, necessary calibration glass, and a housing to enclose the unit in a single package.

3.4.1.1 *Construction (modular).* The requirements for modular construction are not applicable.

3.4.2 *Environmental conditions.* The equipment shall be capable of operation under the

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following conditions as specified in MIL-E-16400:

(a) Temperature, class 4.

(b) Humidity.

3.4.3 *Electrical design.* The nominal supply voltage and frequency shall be 120 volts and 60 cycles per second, single phase. The equipment shall be designed in accordance with the safety requirements of MIL-STD-454.

3.4.3.1 Overload protection shall be provided by fuses but blown fuse indicators are not required. The type of fuseholder shall be type FHN 26G in accordance with MIL-F-19207/16.

3.4.3.2 The equipment shall be grounded in accordance with MIL-E-16400.

3.4.3.3 *Cable entrance.* Cable entrance for power input cable shall be on front panel.

3.4.3.4 External connectors shall be in accordance with MIL-E-16400.

3.4.4 *Maintenance.* Test points and test features are not required.

3.4.5 *Mechanical.*

3.4.5.1 *Shock, vibration and inclination.*

3.4.5.1.1 Only the shock and vibration requirements are applicable to this equipment.

3.4.5.1.2 Shock requirements shall be for grade A, class I, lightweight, type A, of MIL-S-901.

3.4.5.2 The degree of enclosure shall be fogproof, in accordance with MIL-STD-108.

3.4.6 *Controls, indicators and panel layout.*

3.4.6.1 Tactile identification of controls is not required.

3.4.6.2 All controls shall be classified as continually required except voltage selector switch which shall be classified as infrequently required.

3.4.6.3 Locking devices on controls are not required.

3.4.6.4 Elapsed time meters are not required.

3.4.7 Instruction plates shall be mounted on the equipment.

3.5 Performance and detail requirements.

3.5.1 *General description.* Radiac Computer-Indicator CP-95( )/PD shall be an equipment capable of detecting and indicating the amount of X or gamma radiation to which a Radiac Detector DT-60( )/PD has been exposed.

3.5.1.1 *DT-60( )/PD characteristics.* The Radiac Detector DT-60( )/PD is a personnel dosimeter which may be worn about the neck like a pendant, and is constructed in accordance with MIL-D-16153. The X or gamma radiation sensitive volume of this detector is a radiophotoluminescent glass. The glass has the property of luminescing under 3,650-angstrom light after having been exposed to X or gamma radiation. The intensity of the luminescence is proportional to both the dose received and the intensity of the ultraviolet light source. The detector is capable of indicating dosage from 0-1000 roentgens.

3.5.1.2 The luminescent light shall be detected by a photomultiplier tube. This light is in the 4,200 to 7,000 angstrom region of the spectrum. All light except that from the glass shall be excluded from the photomultiplier tube whether the equipment is operated inside or outside the case.

3.5.1.3 The indicating circuitry shall include a blocking capacitor accepting only the alternating current photomultiplier sig-

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nal, in conjunction with amplification, feedback, and filtering circuit components which in combination shall insure linearity and reliability. This indicating circuitry shall be capable of counteracting any long-term phosphorescent effects of contaminants to the DT–60( )/PD glass, which might be observable in addition to the regular silver fluorescence, and shall restrict readings of DT–60( )/PD to fluorescence induced by gamma radiation.

3.5.2 *Range and sensitivity.* The computer-indicator shall operate satisfactorily and in accordance with the requirements set forth in this specification over the range from 0–1000 roentgens This range shall be covered in two steps; namely, 0 to 200 roentgens in 10-roentgen steps and 0 to 1000 roentgens in 20-roentgen steps with each 100-roentgen being a major division.

3.5.2.1 *Reading.* A number of DT–60( )/PD dosimeters irradiated to various dosage levels approved by the command or agency concerned shall be read. Indication shall be within ± 5 percent of the calibrated values of those dosimeters. This permitted variation shall be in addition to errors introduced by the meter itself.

3.5.3 *Reproducibility.* When an exposed Radiac Detector DT–60( )/PD is read in different Radiac Computer-Indicators CP–95( )/PD, the variation in reading from one instrument to another shall not exceed ± 5 percent, or 5 roentgens, whichever is greater. This permitted variation shall be in addition to errors introduced by the meter itself.

3.5.4 *Indicating meter.* The visual indicating meter assembly shall consist of an electrical meter calibrated to read in roentgens and divided into two scales. One scale shall be from 0 to 200 roentgens in 10-roentgen steps, and the other scale from 0 to 1000 roentgens in 20-roentgen steps. Provision shall be made for illumination of the meter scale. Illumination shall be kept to the minimum that will permit reading of the scale. An adjustable red pointer shall be built into the housing for setting the calibration point.

3.5.5 *Controls.*

3.5.5.1 *Power.* A switch, which shall perform the following functions, in the order listed, shall be provided with the control knob mounted externally on the front of the equipment:

(a) Power "off".

(b) Power "on".

(c) Start fluorescent exciter lamp.

3.5.5.2 *Range.* A range-selector switch shall be provided to select either 1000 or 200-roentgen range. A potentiometer shall be provided to permit matching of scales when the ranges are switched.

3.5.5.3 *Calibration control.* A calibration-control knob shall be located on the instrument in an accessible position. Means shall be provided for placing a calibration glass between the ultraviolet-light source and the photomultiplier tube. The calibration point shall be indicated by an adjustable pointer built into the meter housing. The pointer shall be individually adjusted to the value of the calibration standard (calibration glass) for each equipment, and the value of each standard shall be marked on the standard itself.

3.5.5.4 *Manual controls.* Manual controls or switches shall be provided to perform the following functions:

(a) Energize the equipment and start the fluorescent exciter lamp.

(b) Zero the equipment.

(c) Set the calibration.

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(d) Select the range.

3.5.5.5 *Zero and calibration settings.* Prior to determining the dose received by a DT–60( )/PD, the zero and calibration points shall be established. The zero point shall be set by excluding all light from the photomultipliers. The calibration point shall be set by introducing a calibration glass in the reading position. The equipment shall be designed to reduce to a minimum the necessity for setting the calibration and zero points when reading DT–60( )/PD's.

3.5.5.6 *Calibration glass.* The calibration glass shall be equivalent in fluorescent properties to the glass in a DT–60( )/PD which has been exposed to a known dose of X or gamma radiation, except that the fluorescence shall not be affected by exposure to visible, ultraviolet, X, or gamma radiation. The luminescence of the calibration glass shall be equivalent to a dose of 150 to 200 roentgens, unless otherwise agreed upon by the contractor and the command or agency concerned.

3.5.5.7 *Detector positioning assembly.* The detector positioning assembly shall reflect the utmost in simplicity and ease of operation. It shall accommodate all types of DT–60's in the Military system without binding or jamming. The detector positioning assembly shall meet the operational requirements of the specification after 50,000 complete cycles of the assembly.

3.5.5.7.1 The detector positioning assembly shall be so constructed that the operator can calibrate, zero, and read dosimeters in a logical sequence. A positive method to properly align the dosimeter and the calibration standard shall be employed

3.5.6 Operation of the equipment shall be possible after 3 minutes of warm-up time. Full operating stability shall be obtained in 5 minutes of warm-up time.

3.5.7 *Case.* The enclosure shall be of lightweight metal and designed to contain all the computer-indicator parts. The front shall contain the meter face and control switches. Means shall be provided for inserting the DT–60( )/PD for reading the roentgen dosage This entrance shall be light-tight. Provision shall be made for easy access to the interior for servicing the equipment. Provision shall be made to clip a spare photomultiplier and a spare ultraviolet source to the inside of the case. A detachable metal cover shall be provided to cover the front face of the equipment. Means shall be provided within the front cover for storing the power card.

3.5.8 *Size and weight.* The case of lightweight metal, shall not exceed 0.85 cubic feet in size and the computer indicator shall not exceed 23 pounds in weight.

3.5.9 *Carrying handle.* A carrying handle shall be provided on top of the housing and positioned at a point such that the instrument balances when carried. The handle shall be subject to approval by the command or agency concerned.

3.5.10 *Wrenches for Radiac Detector DT–60( )/PD.* A wrench shall be mounted on the outside case of the Radiac Computer-Indicator CP–95( )/PD housing to open the detectors prior to reading in the computer-indicator. In addition, two portable metal wrenches for opening the radiac detector DT–60( )/PD shall be provided. These wrenches shall be held inside the front cover by a clip and secured by a beaded chain or other method approved by the command or agency concerned. The chain shall be so arranged that the wrenches with beaded chain can easily be removed. A third additional wrench capable of seizing the DT–60( )/PD case shall be formed as an identation in the case cover.

3.6 **Painting.** The type of enamel (air-drying speed) shall be optional

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3.6.1 The color shall be as specified for portable shipboard equipment or Marine Corp equipment if specified (see 6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Contractor's quality assurance system.* The contractor shall provide and maintain an effective inspection and quality assurance system acceptable to the Government covering the supplies under the contract The contractor will not be restricted to the inspection station or to the method of inspection listed, provided that an equivalent control is included in the approved quality assurance procedure The contractor shall notify the Government of and obtain approval for any change to the system that might affect the degree of assurance required by this specification or other applicable documents referenced therein

4.2 General inspection. The general inspection of the equipment shall consist of the following classification:

(a) Preproduction inspection.

(b) Production inspection.

(c) Production control inspection.

(d) Environmental tests.

4.2.1 *Test, test equipment and test procedures.* The list of proposed tests, test equipment, and test procedures shall be furnished to the command or agency concerned 60 days or more prior to setting up test equipment.

4.2.2 *Reliability testing.* Reliability testing shall be performed in accordance with test level 2 of MIL–R–22732.

4.2.2.1 Accelerated life tests specified in MIL–E–16400 will not be required for preproduction equipments. In lieu of this test, reliability tests as specified herein will be required.

4.2.2.2 Demonstration of the reliability level of 5,000 hours (MTBF) shall be accomplished by demonstrating a minimum MTBF of 300 hours with 90 percent confidence. The test time on each equipment shall be no more than 700 hours. Accept-reject criteria shall be in accordance with the accept-reject criteria specified in MIL–R–22732.

4.2.2.2.1 Detector positioning assembly life test shall consist of 50,000 complete cycles.

4.2.2.3 Sampling for the reliability tests shall be on the basis of one unit for every 50 units.

4.2.2.4 The performance parameters to be monitored shall be as specified in 3 5.2.

4.3 Preproduction inspection. Preproduction inspection shall consist of the examination and tests specified in 4 3.1 and 4.3.1.1.

4.3.1 The preproduction inspection shall consist of the following tests of MIL–E–16400:

(a) Surface examination.

(b) Operating tests.

(c) Weight and dimensions.

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(d) Supply line voltage and frequency.

(e) Enclosure.

(f) Temperature.

(g) Humidity.

(h) Power.

(i) Shock and vibration.

(j) Radio interference and undesired radiation.

(k) Salt spray test.

(l) Heat test.

4.3.1.1 In addition the following tests shall be performed.

(a) Accuracy (see 3.5.2.1).

(b) Detector positioning assembly life test (see 3.5.5.7).

4.4 Production inspection. Production inspection shall consist of the tests specified in MIL–E–16400, and in addition, shall include the accuracy test of 3.5.2.1.

4.5 Production control inspection. The production control inspection shall consist of the following tests of MIL–E–16400:

(a) Weights and dimensions.

(b) Supply line voltage and frequency.

(c) Heat test.

4.5.1 *Sampling plan.* The unit used for the production control inspection shall be selected from the units passing the production inspection on the basis of one per every 50 units.

4.6 Environmental tests. Environmental tests shall consist of the following tests specified in MIL–E–16400:

(a) Temperature.

(b) Humidity.

(c) Shock and vibration.

4.6.1 *Sampling plan.* The unit used for the environmental test shall be selected from the units passing the production inspection on the basis of one per every 50 units. The environmental tests may be performed concurrently with production control inspection on different samples.

4.7 Disposition of shock tested equipment. Shock tested samples shall be thoroughly reconditioned by the contractor before resubmitting it as production equipment

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts.

5.1.1 *Basic equipment or item.*

5.1.1.1 *Preservation and packaging.* Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the suppliers commercial practice when such meets these requirements.

5.1.1.2 *Packing.* Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice when such meets these requirements.

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5.1.1.3 *Marking.* Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 *Onboard repair parts.* Onboard repair parts shall be preserved, packaged, packed and marked in accordance with MIL–E–17555.

5.2 Domestic shipment and storage of overseas shipment. The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 *Preservation, packaging, packing and marking.* The equipment shall be preserved, packaged, packed and marked for the levels of shipment as specified (see 6.1) in accordance with MIL–E–17555)

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Color requirement (see 3.4.1.2.1).

(c) Preservation, packaging, packing and marking requirements other than those required by 5.1 (see 5.2).

6.2 Preproduction.

6.2.1 Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.3 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE

Custodians:
Army — EL
Navy — SH
Air Force — 11

Review activities:
Army — MI, EL
Navy — SH, WP, YD
Air Force — 11, 15

User activities:
Army — MU
Navy — MC
Air Force — 17

Preparing activity:
Navy — SH
Project No. 6665-0125

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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